

Introduction

SGA 2, Grothendieck's introduction to SGA 2

English draft

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About this draft

This is an unofficial English translation of Grothendieck's Introduction, from SGA 2. The source volume is A. Grothendieck (notes by a group of auditors), with an exposé by M. Raynaud, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux* (SGA 2), Séminaire de géométrie algébrique du Bois Marie, 1962. It follows the slightly corrected SMF recomposition, arXiv:math/0511279.

This draft contains the assigned body of the source, including proofs, original footnotes, and editor notes (N.D.E.). Statement numbering and mathematical notation follow the source. References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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Introduction

We present here, in revised and completed form, a photo-offset re-edition of the second Seminar of Algebraic Geometry of the Institut des Hautes Études Scientifiques held in 1962 (mimeographed).

The reader is referred to the Introduction to the first of these Seminars (cited SGA 1 below) for the aims pursued by these seminars, and their relations with the *Éléments de Géométrie Algébrique*.

The text of Exposés I to XI was written as we went along, from my oral exposés and manuscript notes, by a group of auditors, comprising I. Giorgiutti, J. Giraud, Mlle M. Jaffe (who became Mme M. Hakim) and A. Laudal. These notes were originally regarded as provisional and of very limited circulation, pending their absorption into the EGA (an absorption which has now become at the very least problematic, as for the other parts of the SGA). As was said in the notice to the original edition, this “confidential” character of the notes was to excuse certain “weaknesses of style”, doubtless more apparent in the present Seminar SGA 2 than in the others. I have tried as far as possible to remedy this in the present re-edition, by a relatively close revision of the initial text. In particular I have harmonized the numbering systems of the statements used in the different exposés, by introducing everywhere the same decimal system, already used in most of the original exposés of the present SGA 2, as well as in all the other parts of the SGA. This led me in particular to revise entirely the numbering¹ of the statements of Exposés III to VIII (and consequently, of the references to those exposés)². I have also tried to extirpate from the original text the principal typing or syntax errors (which were numerous and troublesome). Moreover, Mme M. Hakim kindly undertook to rewrite Exposé IV in a less telegraphic style than the original exposé. As in the other re-editions of the SGA, I have also added a certain number of footnotes, either to give additional references, or to indicate the state of a question for which progress has been made since the writing of the original text. Finally, this Seminar has been augmented by a new exposé, namely Exposé XIV, written by Mme Michèle Raynaud in 1967, which takes up and completes suggestions contained in the “Comments on Exposé XIII” (XIII XIII.6) (written in March 1963). This exposé takes up the Lefschetz-type theorems from the point of view of étale cohomology, using the results on étale cohomology set out in SGA 4 and SGA 5 (to appear in this same collection *Series in Pure Mathematics*)³; it is therefore in this respect of a less “elementary” nature than the other exposés of the present volume, which use little more than the substance of Chapters I to III of EGA. Here is a sketch of the contents of the present volume. Exposé I contains the sorites of “cohomology with support in Y ” $H_Y^*(X, F)$, where Y is a closed subset of a space X , a cohomology which can be interpreted as a cohomology of X modulo the open set $X - Y$, and which is the abutment of a most useful “*spectral sequence of passage from local to global*” I I.2.6, involving *sheaves* of cohomology “with support in Y ”

¹**N.D.E.** one has preserved the original numbering as far as possible, adding the adverb *bis* when ambiguous duplicates occurred here and there.

²It goes without saying that all references to SGA 2 appearing in the parts of the SGA published in the *Series in Pure Mathematics* will refer to the present volume, and not to the original edition of SGA 2!

³**N.D.E.** in fact these seminars are published by Springer (numbers 269, 279, 305 and 589), but, alas, are out of print.

$H_Y^*(F)$ ⁴. This formalism can in many questions play a role of “localization” analogous to that played by the consideration of “tubular” neighbourhoods of Y in differential geometry. Exposé II studies the preceding notions in the case of quasi-coherent sheaves on preschemes; Exposé III gives their relation with the classical notion of *depth* (III III.3.3).

Exposés IV and V give notions of *local duality*, which one may compare with Serre’s projective duality theorem (XII XII.1.1); we note that these two types of duality theorems are generalized in a substantial way in Hartshorne’s seminar (cited in a footnote at the end of Exp. IV)⁵

Exposés VI and VII give easy technical notions, used in Exposé VIII to prove the *finiteness theorem* (VIII VIII.2.3), giving necessary and sufficient conditions, for a coherent sheaf F on a noetherian scheme X , in order that the local cohomology sheaves $H_Y^i(F)$ be coherent for $i \leq n$ (or equivalently, that the sheaves $R^i f_*(F|X - Y)$ be coherent for $i \leq n - 1$, where $f: X - Y \rightarrow X$ is the inclusion). This theorem is one of the central technical results of the Seminar, and we show in Exposé IX how a theorem of this nature can be used to establish a “comparison theorem” and an “existence theorem” in formal geometry, following and generalizing the use made in (EGA III §§4 and 5) of the finiteness theorem for a proper morphism.

One applies these last results in X and XI, devoted respectively to Lefschetz-type theorems for the fundamental group, and for the Picard group. These theorems consist in comparing, under certain conditions, the invariants (π_1 or Pic) attached respectively to a scheme X and to a subscheme Y (playing the role of a hyperplane section), and in particular in giving conditions under which they are isomorphic. Roughly speaking, the hypotheses made serve to pass from Y to the formal completion of X along Y , and then to be able to apply the results of IX to pass from there to an open *neighbourhood* U of Y in X . To pass from U to X , one still needs information (of “purity” or “parafactoriality” type) for the local rings of X at the points of $Z = X - U$ (which is a finite discrete set in the cases envisaged). This explains the interaction, in the proofs of Exposés X, XI, XII, between local and global results, notably in certain inductions. The principal results obtained in X and XI are the theorems of *local nature* X X.3.4 (*purity theorem*) and XI XI.3.14 (*parafactoriality theorem*). One will note that these theorems are proved by cohomological techniques, of an essentially global nature. In XII, using the preceding local results, one obtains the global variants of these results for projective schemes over a field, or more generally over a more or less arbitrary base scheme; among the typical statements, we mention XII XII.3.5 and XII XII.3.7.

In XIII, we review some of the numerous problems and conjectures suggested by the results and methods of the Seminar. The most interesting, perhaps, concern the cohomological and homotopical Lefschetz-type theorems for complex analytic spaces, cf. XIII pages 26 and following⁶. In the context of the étale cohomology of schemes, the corresponding conjectures are proved in XIV by a duality technique which should apply equally in the complex analytic case (cf. comments XIII p. 25 and XIV XIV.6.4). But the corresponding

⁴**N.D.E.** one has retained the underlined notations for the sheafified versions of functors, the calligraphic analogue of Γ not being clear.

⁵**N.D.E.** Hartshorne’s book contains sign errors and, above all, does not really prove the compatibility of the trace with change of base. Conrad has taken this work up completely, proving this compatibility, which is crucial and highly non-trivial (*Grothendieck duality and base change*, Lect. Notes in Math., vol. 1750, Springer-Verlag, Berlin, 2000). Alas, errors remain (cf. two preprints (Conrad B., “Clarifications and corrections to “*Grothendieck duality and base change*””) and “An addendum to Chapter 5 of “*Grothendieck duality and base change*””). For a more concrete aspect, with particular attention to the notion of residue, see the works of Lipman, in particular (Lipman J., *Dualizing sheaves, differentials and residues on algebraic varieties*, Astérisque, vol. 117, Société mathématique de France, 1984). A categorical proof of the duality theorem, based on Brown’s representability theorem, was obtained by Neeman (Neeman A., “The Grothendieck duality theorem via Bousfield’s techniques and Brown representability”, *J. Amer. Math. Soc.* **9** (1996), no. 1, p. 205–236).

⁶**N.D.E.** essentially all the conjectures stated in XIII and XIV are now proved; see the footnotes of those sections for references and comments.

homotopical statements in the case of analytic spaces (and more particularly the statements involving the fundamental group) seem to require entirely new techniques (cf. XIV XIV.6.4).

I am happy to thank all those who, in various capacities, have helped with the appearance of the present volume, including the collaborators already cited in this Introduction. More particularly, I wish to thank Mlle Chardon for the good grace with which she discharged the thankless task constituted by the material preparation of the final manuscript for photo-offset.

Bures-sur-Yvette, April 1968

A. Grothendieck.